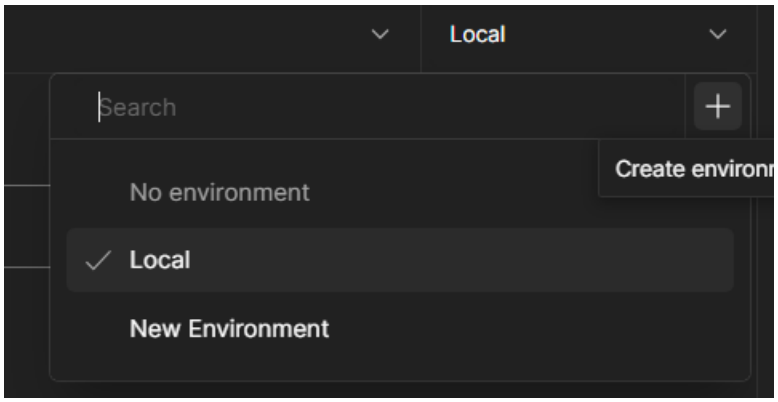
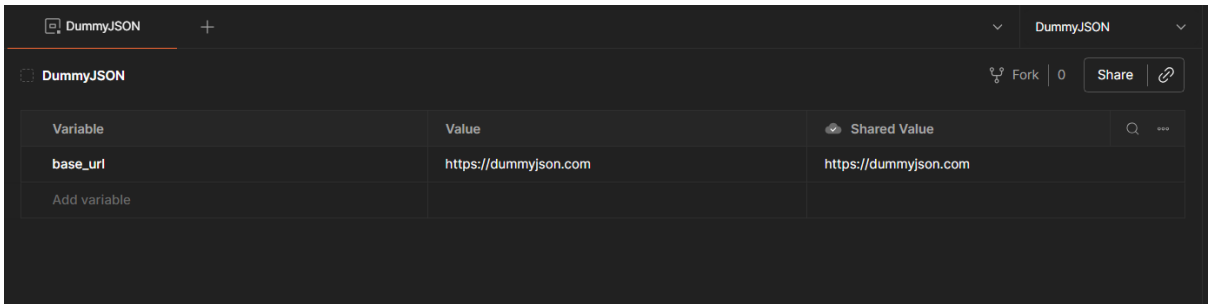
E-commerce Backend API Testing (Postman)

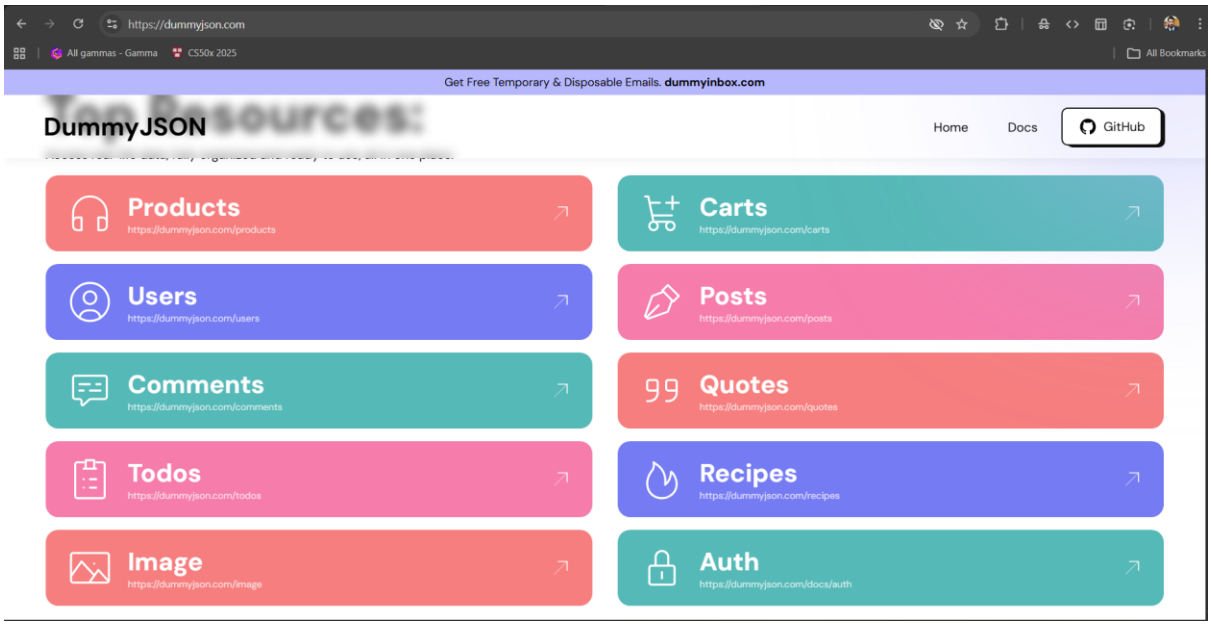
Create a new Environment, Name = DummyJSON

base_url = https://dummyjson.com





Feature	*Endpoint*
Login	/auth/login
Users	/users
Products	/products
Cart	/carts/user/1
Single product	/products/1
Search	/products/search?q=phone



Name Post: {{base_url}}/auth/login

Save it on New Collection: DummyJSON

Authentication: -

Create request: POST {{base_url}}/auth/login

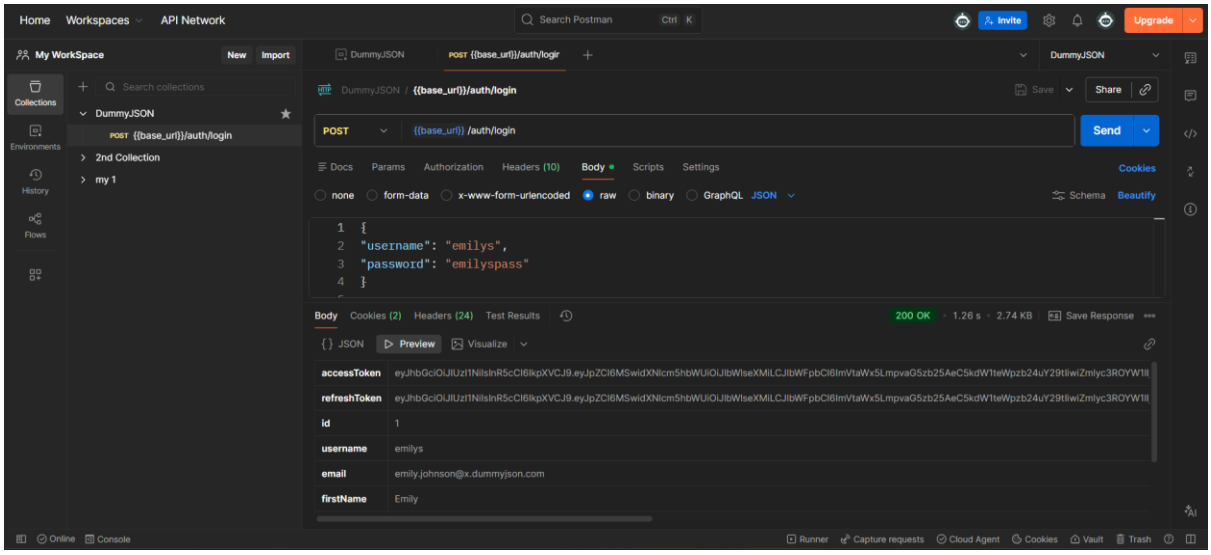
Body → raw → JSON:

{

"username": "emilys",

"password": "emilyspass"

}

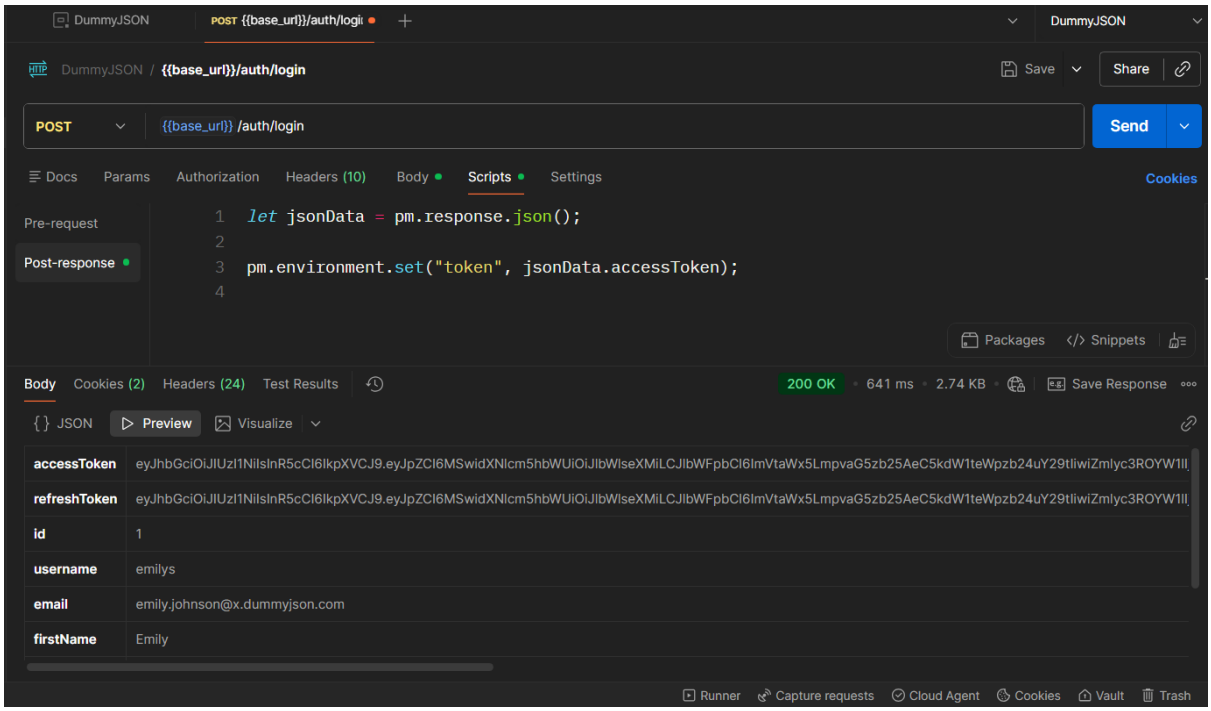


Save the Token Automatically: -

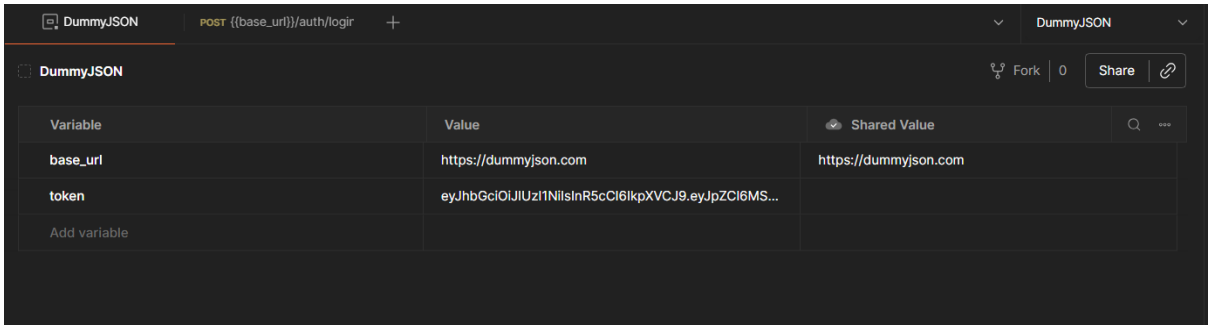
Open your **Login request**
Go to **Tests tab** (not Pre-request)

let jsonData = pm.response.json();

pm.environment.set("token", jsonData.accessToken);



Prove the token is saved: -



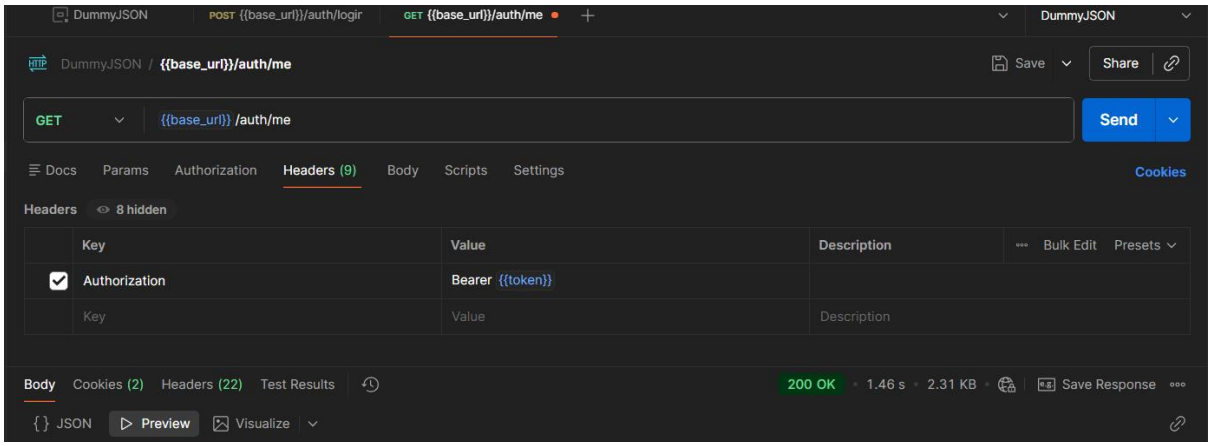
AUTHENTICATION TESTING: -

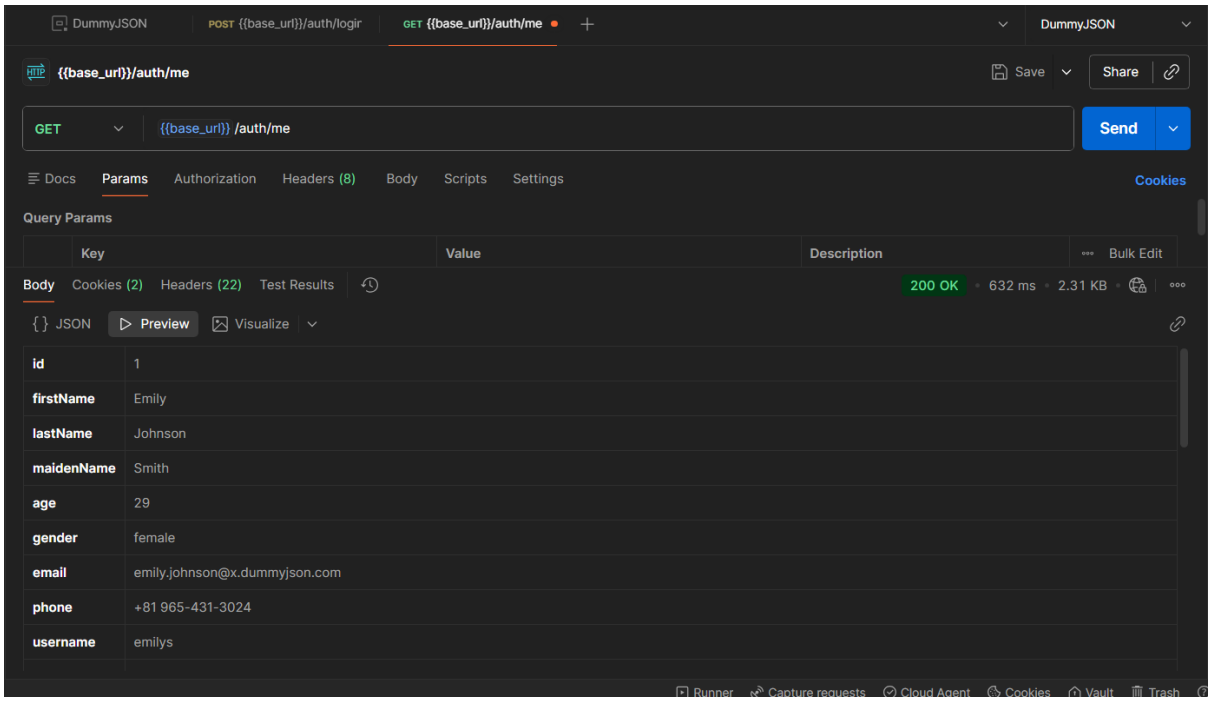
GET {{base_url}}/auth/me

Headers tab:

Key: Authorization

Value: Bearer {{token}}





Add the automation: - put it at **collection level**.

Do this:

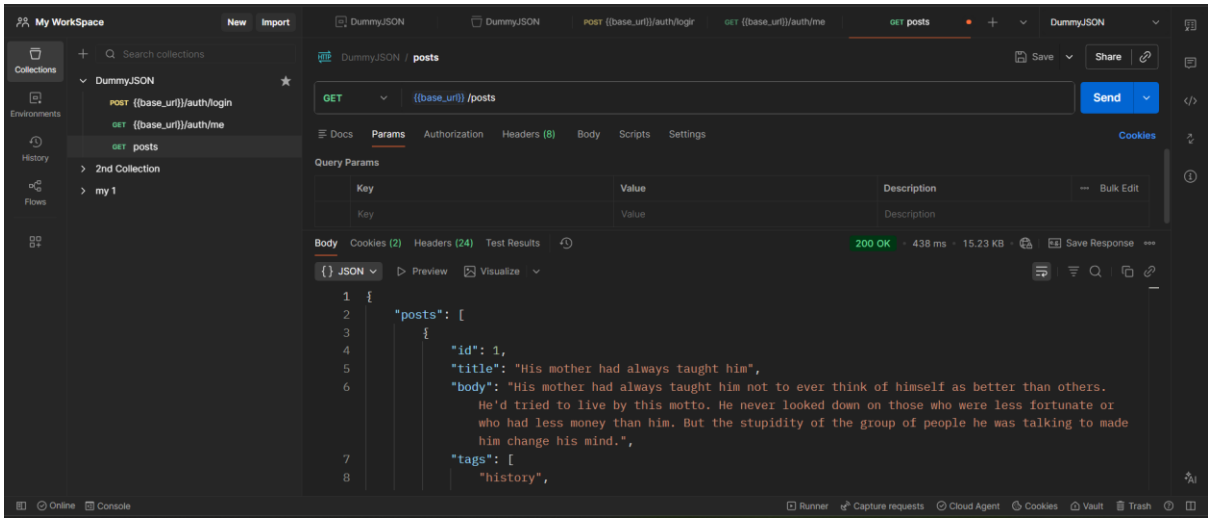
Right-click → DummyJSON → Edit → Pre-request Script

```
// Auto login if token not available
if (!pm.environment.get("token")) {
pm.sendRequest({
  url: pm.environment.get("base_url") + "/auth/login",
  method: "POST",
  header: {
    "Content-Type": "application/json"
  },
  body: {
    mode: "raw",
    raw: JSON.stringify({
      username: "emilys",
      password: "emilyspass"
    })
  }
}, function (err, res) {
  let json = res.json();
  pm.environment.set("token", json.accessToken);
});
}
```

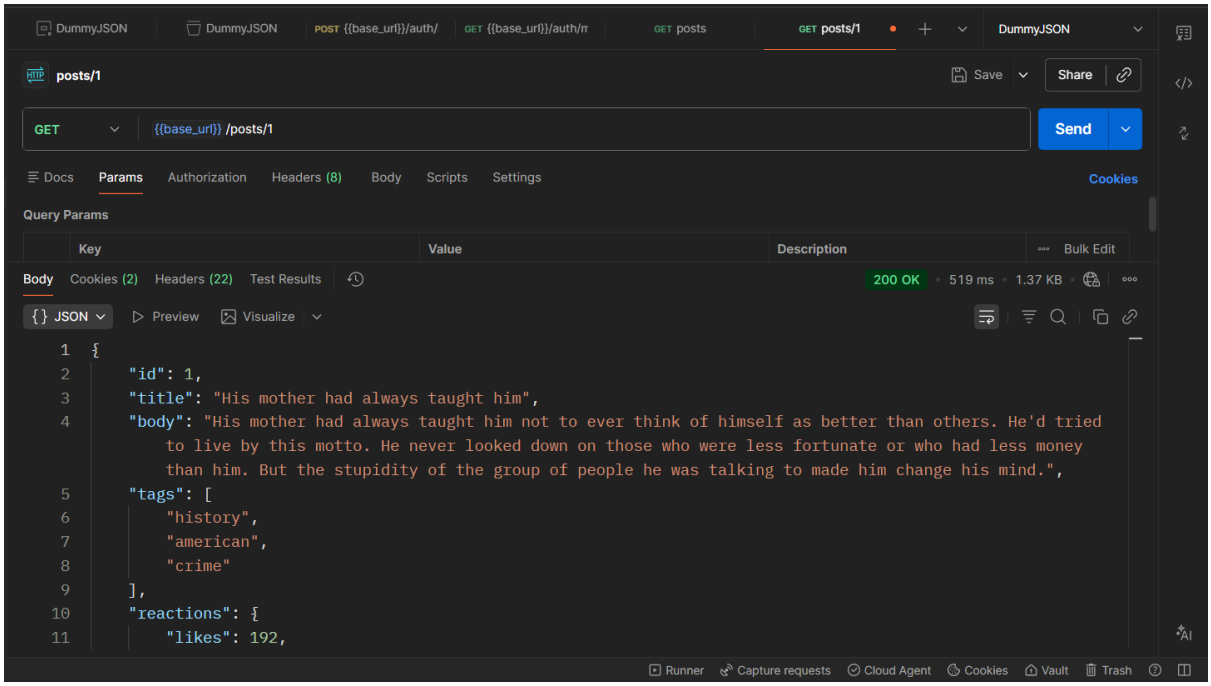
```
// Auto login if token not available
if (!pm.environment.get("token")) {

```
pm.sendRequest({
    url: pm.environment.get("base_url") + "/auth/login",
    method: "POST",
    header: {
        "Content-Type": "application/json"
    },
    body: {
        mode: "raw",
        raw: JSON.stringify({
            username: "emilys",
            password: "emilyspass"
        })
    }
}, function (err, res) {
    let json = res.json();
    pm.environment.set("token", json.accessToken);
});
```
```

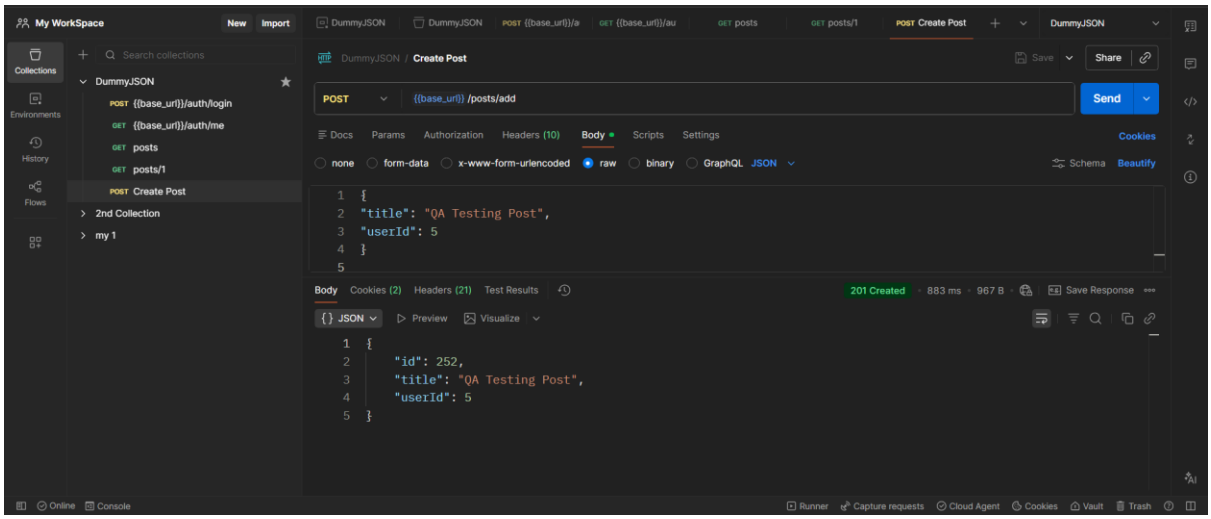
GET post: -



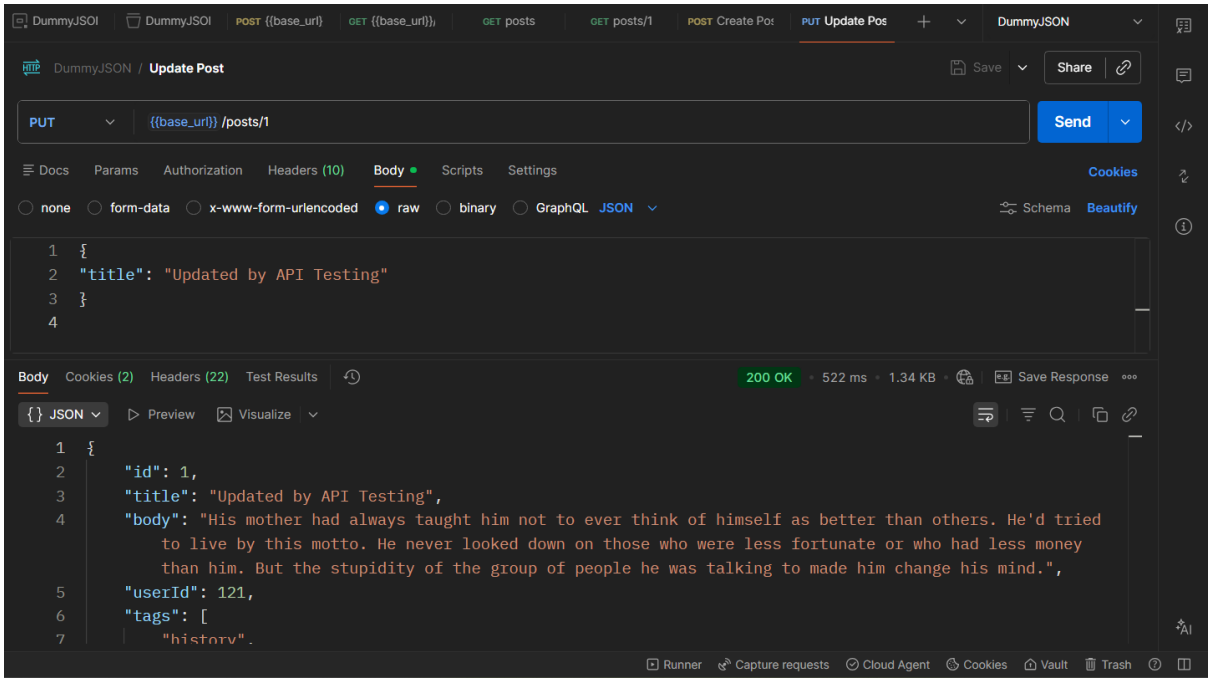
Get single post: -



Create Post: -



Update Post: -


DummyJSON
DummyJSON
POST {{base_url}
GET {{base_url}}
GET posts
GET posts/1
POST Create Pos
PUT Update Pos
DummyJSON
DummyJSON / Update Post
Save
Share
PUT
{{base_url}} /posts/1
Send
Docs
Params
Authorization
Headers (10)
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"title": "Updated by API Testing"
}
Body
Cookies (2)
Headers (22)
Test Results
200 OK
522 ms
1.34 KB
Save Response
JSON
Preview
Visualize
{
"id": 1,
"title": "Updated by API Testing",
"body": "His mother had always taught him not to ever think of himself as better than others. He'd tried to live by this motto. He never looked down on those who were less fortunate or who had less money than him. But the stupidity of the group of people he was talking to made him change his mind.",
"userId": 121,
"tags": [
"history",
Runner
Capture requests
Cloud Agent
Cookies
Vault
Trash

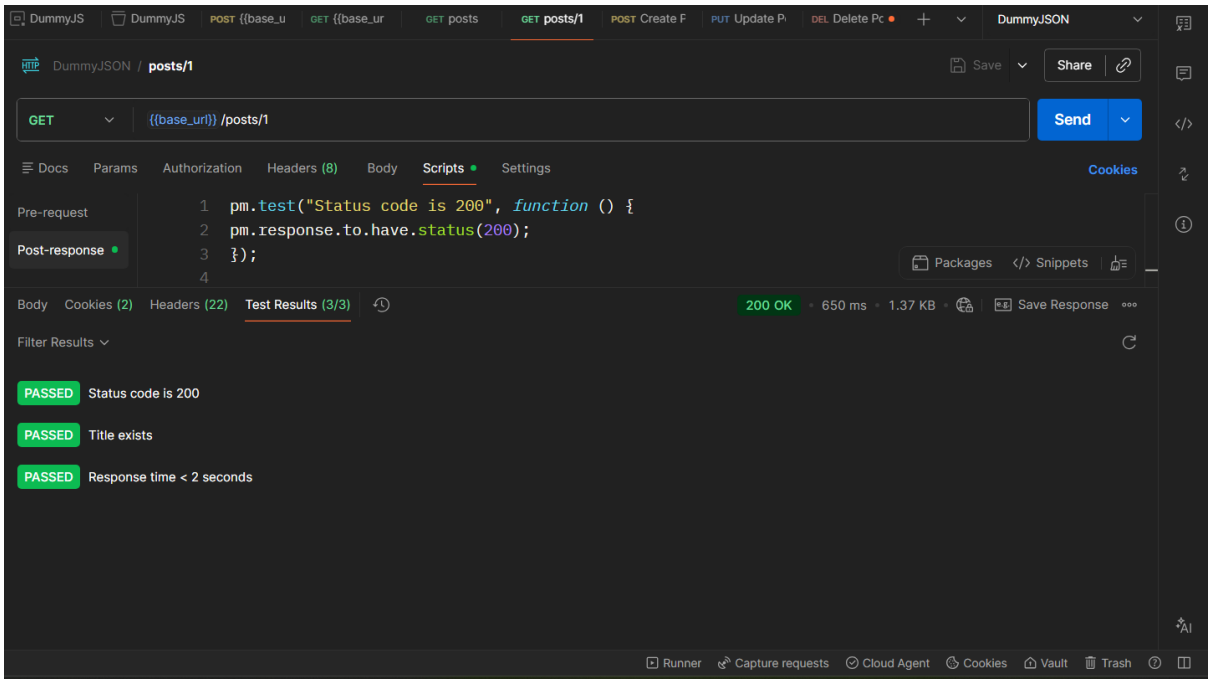
Test: -

Inside **GET /posts/1 → Tests tab**:

```
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});

pm.test("Title exists", function () {
let data = pm.response.json();
pm.expect(data.title).to.exist;
});

pm.test("Response time < 2 seconds", function () {
pm.expect(pm.response.responseTime).to.be.below(2000);
});
```



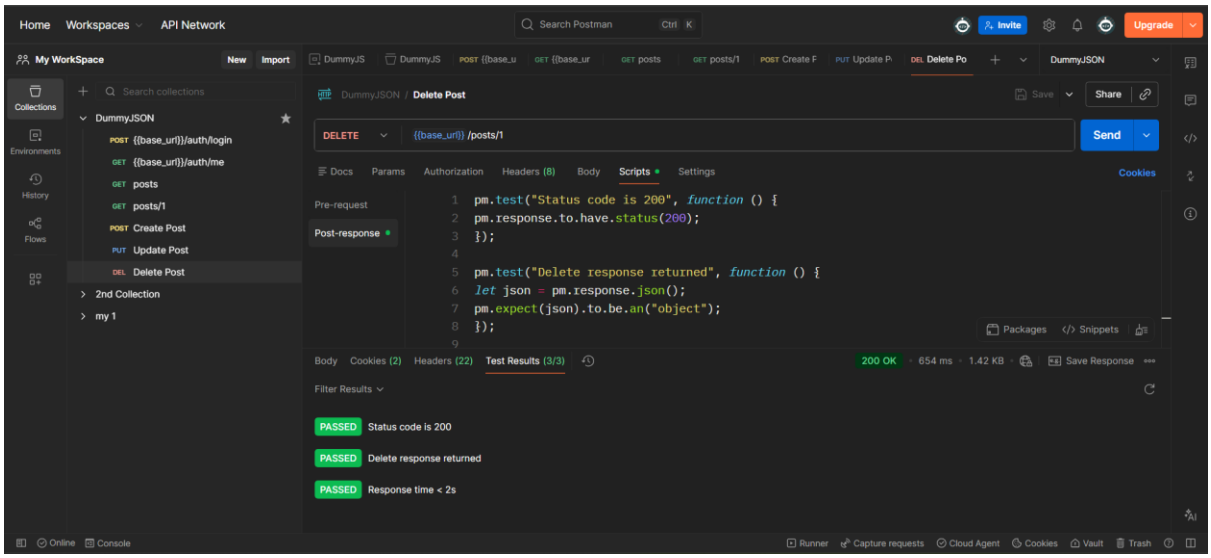
Delete: -

Open your DELETE request → **Tests tab**

```
pm.test("Status code is 200", function () {
pm.response.to.have.status(200);
});

pm.test("Delete response returned", function () {
let json = pm.response.json();
pm.expect(json).to.be.an("object");
});

pm.test("Response time < 2s", function () {
pm.expect(pm.response.responseTime).to.be.below(2000);
});
```



Negative Test: - (knowingly use false info)

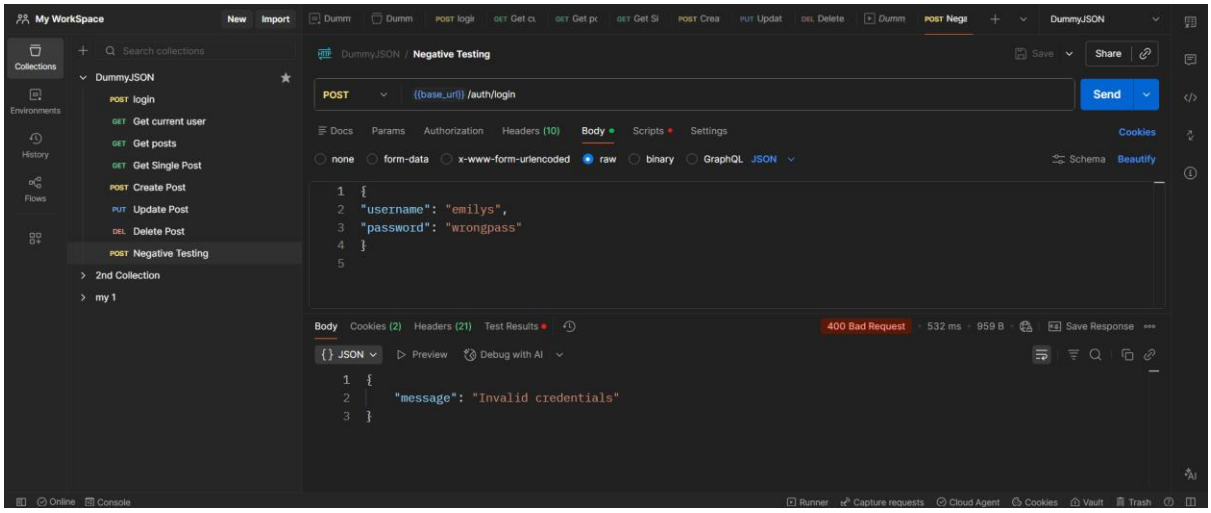
POST {{base_url}}/auth/login

Body:

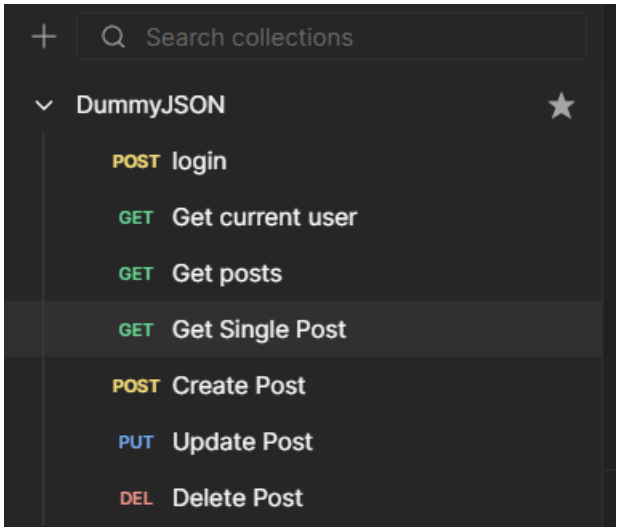
```
{
"username": "emilys",
"password": "wrongpass"
}
```

Test tab:

```
pm.test("Login should fail", function () {
pm.response.to.have.status(400);
});
```



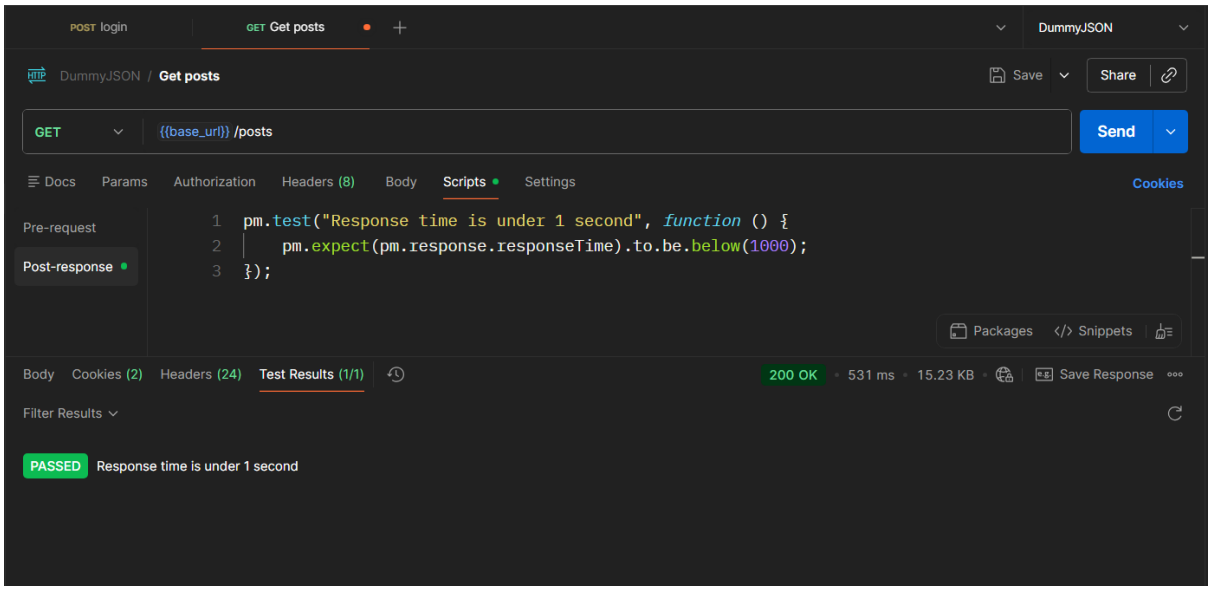
Name change Collection: -



Performance Testing: -

Open **any request(GET posts) → Tests**

```
pm.test("Response time is under 1 second", function () {
    pm.expect(pm.response.responseTime).to.be.below(1000);
});
```



Functional Testing (Status Code + Data Validation): -

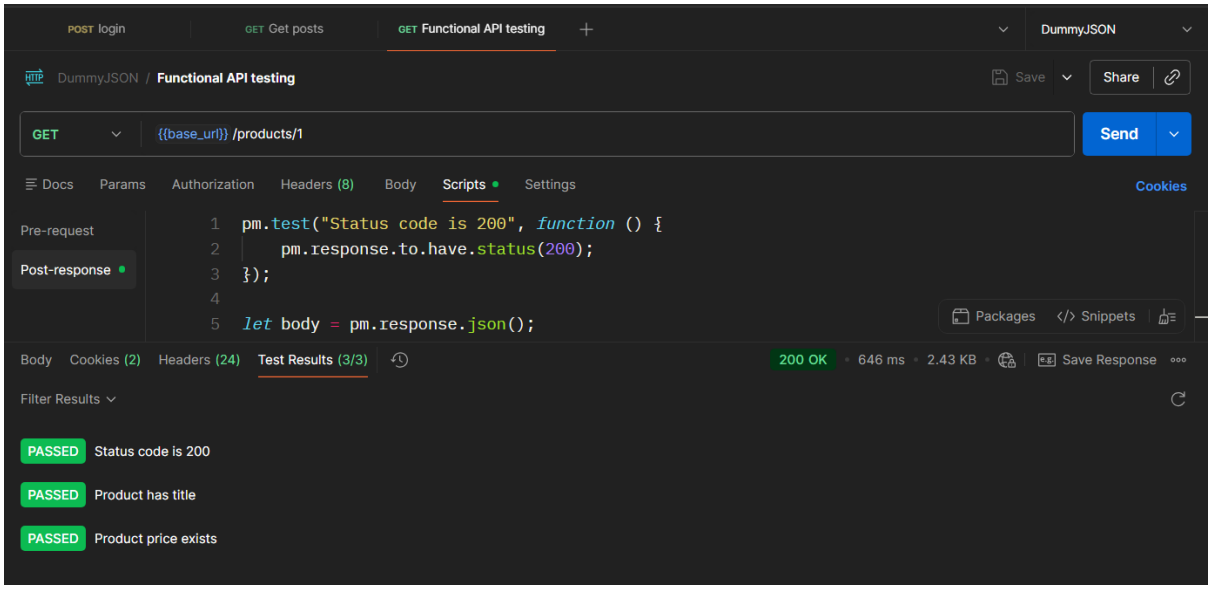
GET {{base_url}}/products/1 -> Test

```
pm.test("Status code is 200", function () {
   pm.response.to.have.status(200);
});

let body = pm.response.json();

pm.test("Product has title", function () {
   pm.expect(body.title).to.be.a("string");
});

pm.test("Product price exists", function () {
   pm.expect(body.price).to.exist;
});
```

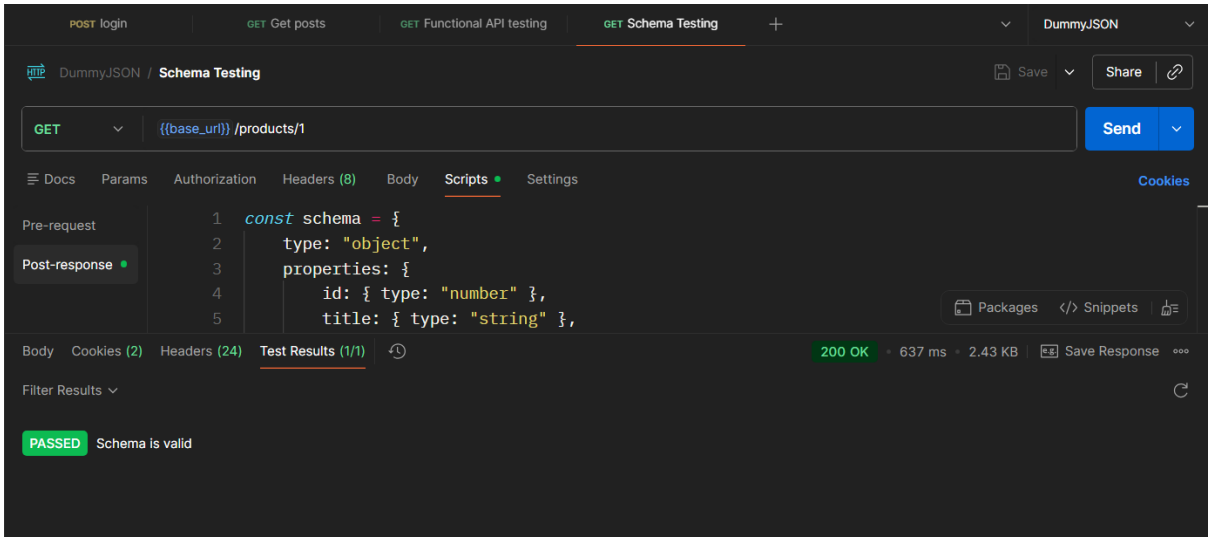


Schema Validation: -

GET {{base_url}}/products/1 -> Test

```
const schema = {
   type: "object",
   properties: {
      id: { type: "number" },
      title: { type: "string" },
      price: { type: "number" }
   },
   required: ["id", "title", "price"]
};

pm.test("Schema is valid", function () {
   pm.expect(tv4.validate(pm.response.json(), schema)).to.be.true;
});
```

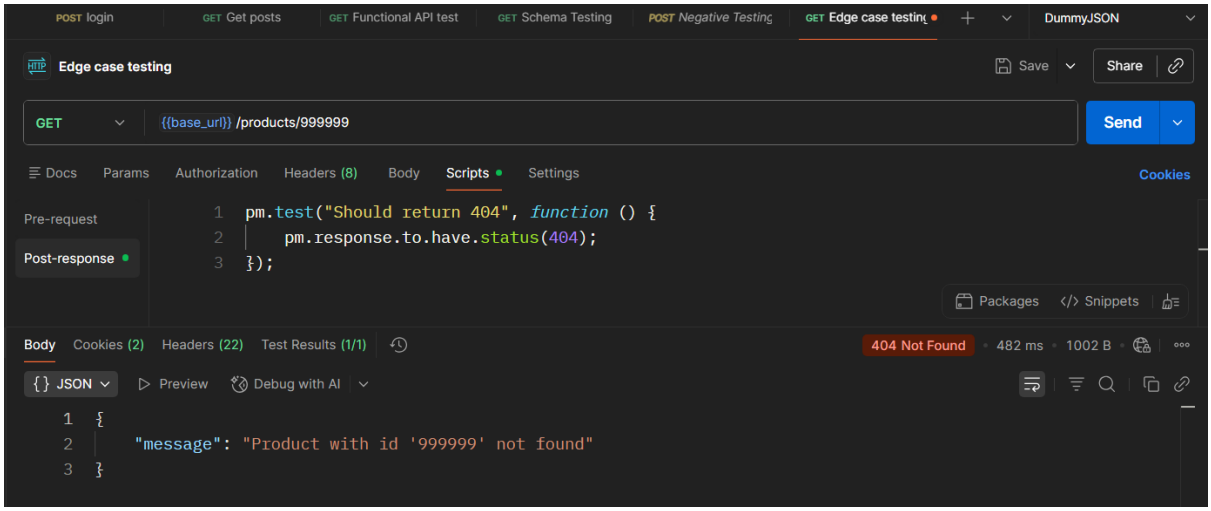


Edge case testing: -

GET {{base_url}}/products/999999

Test: -

```
pm.test("Should return 404", function () {
    pm.response.to.have.status(404);
});
```



Security Testing — Invalid Token: -

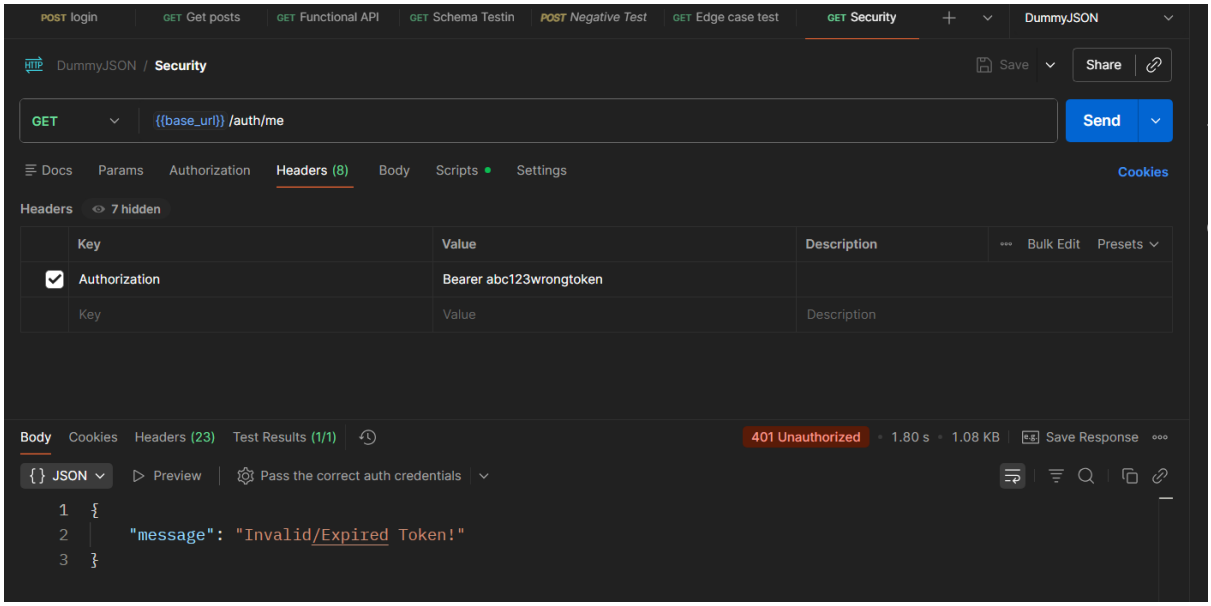
GET {{base_url}}/auth/me

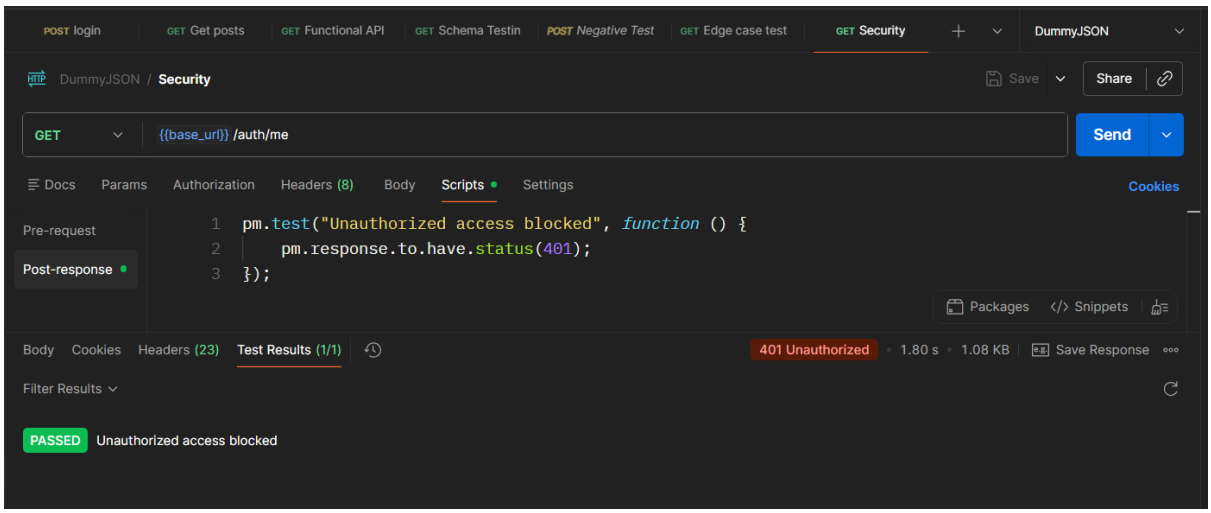
Header:

Authorization: Bearer abc123wrongtoken

Test:

```
pm.test("Unauthorized access blocked", function () {
   pm.response.to.have.status(401);
});
```





Run Collection: -

Top left → click **Run** (beside collection)

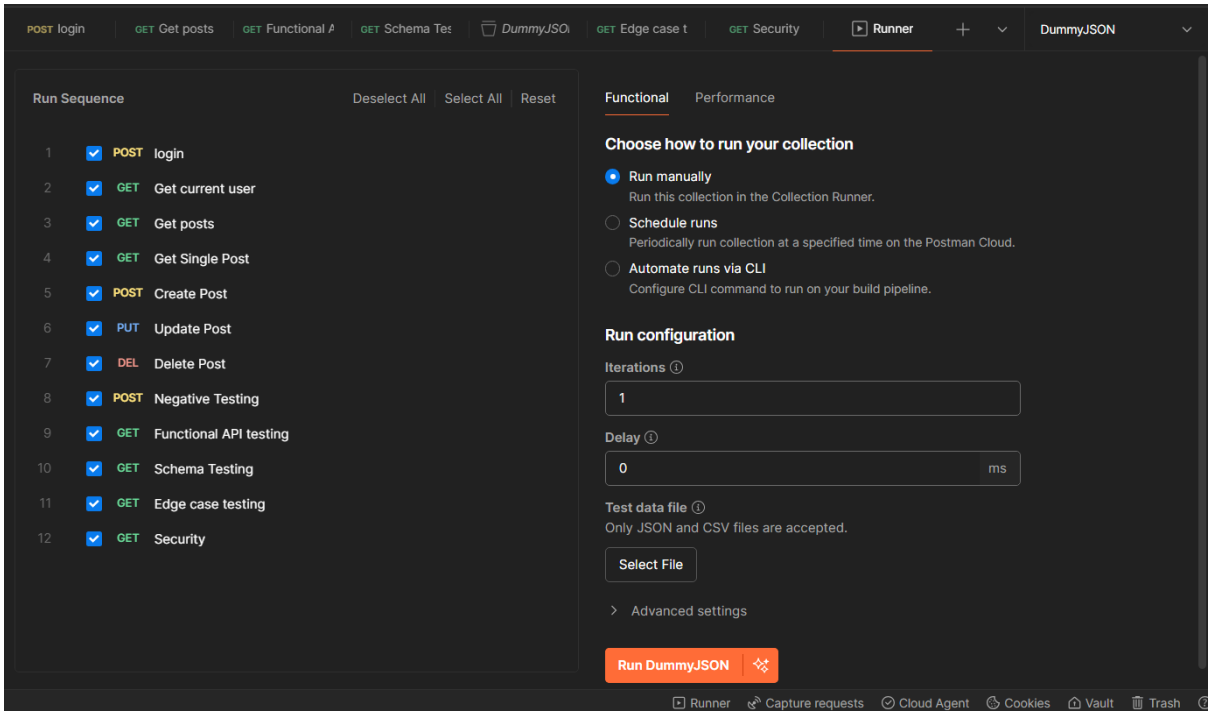
The Collection Runner window opens.

Select:

- Environment: your DummyJSON environment
- Iterations: 1

Then click:

Run DummyJSON



DummyJSON - Run results ERROR

Debug with AI | Run Again | New Run | Automate Run | Share

Ran today at 01:14:28 PM · View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	DummyJSON	1	4s 659ms	7	1	505 ms

All Tests | Passed (7) | Failed (0) | Skipped (0) | Errors (1) | Console Log | View Summary

No tests found

PUT **Update Post**
https://dummyjson.com/posts/1
200 • 405 ms • 1.385 KB •

No tests found

DELETE **Delete Post**
https://dummyjson.com/posts/1
200 • 693 ms • 1.453 KB • 3

PASS Status code is 200

PASS Delete response returned

PASS Response time < 2s

POST **Negative Testing**
https://dummyjson.com/auth/login
400 • 420 ms • 959 B •

No tests found

Runner | Capture requests | Cloud Agent | Cookies | Vault | Trash

POST login | GET Get posts | GET Functional A | GET Schema Tes | POST Negative T | GET Edge case t | GET Security | DummyJSOI | DummyJSON

DummyJSON - Run results ERROR

Debug with AI | Run Again | New Run | Automate Run | Share

Ran today at 01:14:28 PM · View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	DummyJSON	1	4s 659ms	7	1	505 ms

All Tests | Passed (7) | Failed (0) | Skipped (0) | Errors (1) | Console Log | View Summary

Iteration 1

POST **Negative Testing**
https://dummyjson.com/auth/login
400 • 420 ms • 959 B •

SCRIPT Post-response: SyntaxError: Unexpected identifier 'tab'